

Question	Marks
1(a) Difference in pressure $\Delta p = \rho g \Delta h$ $= 1000 \times 9.81 \times 0.051$ $= 500.31 \text{ Pa}$ $= 500 \text{ Pa}$	M1 A1
1(b) Upthrust $U = \Delta p \times 5.1 \times 5.1 \times 10^{-4}$ $= 1.3005 = 1.3 \text{ N}$	M1 A0
1(c) $U + T = W$ $1.3 + T = 4.0$ $T = 2.7 \text{ N}$ By N3L Force on spring by cube is 2.7 N downwards.	M1 A1
1(d) Let the extension of the spring be x . $T = k x$ $2.7 = 30 x$ $x = 0.090 \text{ m} = 9.0 \text{ cm}$ initial height above surface, h	M1 A0 M1
	$= 9.0 - 7.0$ $= 2.0 \text{ cm}$ $= 0.020 \text{ m}$ A1
1(e)(i) $2.7 = 4.0/9.81 \times a$ $a = 6.62 \text{ m s}^{-2}$ (Downwards)	M1 A1
1(e)(ii) Viscous force is proportional to speed. Resultant force downwards = Weight – Viscous Force (- Upthrust if considered not negligible) a decreases as velocity increases.	B1
When viscous force = weight, a = 0 m s ⁻²	B1

Question

2(a)

acceleration / force proportional to displacement (from a fixed point)

Marks

B1